



Sijunzi Decoction treatment improved depression with qi deficiency/fatigue in the strain-dependent manner, involved in anti-inflammation in the hippocampus and muscles

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ABSTRACT

Ethnopharmacological relevance: In the traditional Chinese medicine (TCM) clinic, qi deficiency is a common syndrome pattern in major depressive disorder (MDD). Sijunzi decoction (SJZD), a classic multi-herbal formula to replenish qi and nourish blood, is widely used to treat qi deficiency syndrome. The symptoms of qi deficiency are very similar to fatigue. Currently the mainstream antidepressant treatment outcome for depression with fatigue remains unsatisfying. SJZD has potential for improvement in the treatment of depression with qi deficiency, which has not been scientifically characterized previously.

Aim of the study: The study aimed to test the effects of SJZD in a mouse model of depression with qi deficiency/fatigue and to investigate the associated mechanisms, focusing on inflammation in the hippocampus and muscles.

Materials and methods: Balb/c and 129S1/SvImJ (129/S1) strains of mice were compared for depression-like and qi deficiency-like behaviors following receiving the same procedure of chronic unpredictable mild stress (CUMS). Qi deficiency behavior was assessed using grip strength test (GST), exhaustive swimming test (EST) and degree of redness (DOR). Depressive behavior was assessed using sucrose preference test (SPT), tail suspension test (TST), and forced swimming test (FST). SJZD was administered for 1 week in CUMS-exposed mice in both strains. The conventional antidepressant fluoxetine (FLX), ineffective to fatigue/qi deficiency, was used to compared with SJZD, qPCR was used to detect gene expressions of inflammatory factors in the hippocampus and muscles in both strains.

Results: Balb/c and 129/S1 mice both showed depressive symptoms comparably after exposed to CUMS. However, qi deficiency symptoms were only shown in Balb/c mice, with decreased grip strength in GST, reduced swimming times in EST and decreased degree of redness in DOR. SJZD was able to reverse both depressive deficits and qi deficiency in Balb/c mice, without influencing 129/S1 mice. Consistent with the depression-phenotype, the expressions of the inflammatory factors including IL-1 β , TNF- α , NF- κ B and CD8 in the hippocampus were upregulated in both Balb/c and 129/S1 mice, which were reversed by SJZD only in Balb/c mice. Consistent with the qi deficiency-phenotype, the expressions of these inflammatory factors were up-regulated in the muscles only in Balb/c mice, which were reversed by SJZD. In contrast, FLX elicited antidepressant effects

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without changing qi deficiency in Balb/c mice, consistent with its improvement of inflammation in the hippocampus, but not muscles.

Conclusions: Balb/c strain mice showed co-susceptibility to depression and qi deficiency/fatigue following CUMS, both of which were alleviated by SJZD. These effects were associated with the suppression of inflammation in the hippocampus and muscles respectively, suggesting that the anti-inflammation effects of SJZD on both brain and the peripheral systems may play a part in qi-replenishing and antidepressant functions. Our study provides the first scientific evidence, leveraging animal genetics, to demonstrate the necessity and efficacy of stratified treatment of depression, using TCM treatment based on syndrome differentiation.

1. Introduction

Major depressive disorder (MDD) is one of mental illness with high rate of self-harm and suicide, which has brought serious burden to society and families (2020). There are multiple clinical manifestations of depression, including low mood and loss of interest or anhedonia, slow thinking, and fatigue (Wen et al., 2022). In a remarked number of MDD patients, the clinical manifestations are lack of energy, lazy speech and other symptoms of fatigue (Bondar et al., 2020). Conventional antidepressants, such as selective serotonin reuptake inhibitor (SSRI), generally have limited effects on alleviating fatigue symptoms, which could lead to poor adherence to treatment, and increased risk of relapse of depression (Papakostas, 2008). Drugs that alleviate depression while relieving fatigue are expected to have better clinic outcomes for such subtype of MDD patients.

In traditional Chinese medicine (TCM), the diagnosis and treatment are based on Zheng or syndrome pattern, which is different from but related with the process of distinguishing and diagnosing diseases in western medicine (Bondar et al., 2020; Chen et al., 2019; Li et al., 2020; Nerurkar et al., 2019). The typical symptoms of fatigue are overlapped with the syndrome of qi deficiency, mainly manifested as weakness in the limbs and lazy speech (Chen et al., 2019). Some randomized controlled trials reported that qi deficiency/fatigue is often observed at the time of clinical depression (Macpherson et al., 2013; Yeung et al., 2015). Sijunzi decoction (SJZD, a classic qi-tonifying formula) consisted of 4 herbs including *Radix Codonopsis* (Dangshen, dried roots of *Codonopsis pilosula* Nannf.), *Atractylodis Macrocephalae Rhizoma* (Baizhu, dried roots of *Atractylodes macrocephala* Koidz.), *Poria* (Fuling, dried sclerotium of *Poria cocos* (Schw.) Wolf), *Glycyrrhizae Radix Et Rhizoma Praeparata Cum Melle* (Gancao, dried roots and rhizomes of *Glycyrrhiza uralensis* Fisch.) (All herbs information were referred to 'Chinese Pharmacopoeia' and the plant names have been checked with MPNS, <http://mpns.kew.org>). SJZD is capable to replenish qi and nourish blood (Gao et al., 2018; Zhao et al., 2020). Although SJZD is a renowned for the treatment of qi deficiency syndrome in TCM (Guo et al., 2022; Liu, W.J. et al., 2021), the mechanism underlying its effects on depression and qi deficiency remains elusive.

Several factors may commonly influence fatigue and depression, including the dysregulation of HPA axis and inflammation. Studies have shown that fatigue is associated with the release of inflammatory cytokines such as interleukin-1 (IL-1), IL-6, IL-10 and tumor necrosis factor- α (TNF- α) (Dreisbach et al., 1998), etc. Similarly, qi deficiency also has close links with dysregulated inflammation, including IL-6 and IL-10 (Hu et al., 2024). Low grade activation of inflammation in the hippocampus or other brain regions were also observed in chronic stress models of depression as well as in MDD patients, with activation of inflammation in the brain, such as the increase of IL-1 β , TNF- α and CD8 (Sun et al., 2022; Zhu et al., 2023). Although common mechanisms in activation of inflammatory factors appeared to be shared, it is still unclear whether abnormalities in inflammation in central nervous and peripheral systems have distinct roles in simultaneous occurrence of depression and qi deficiency/fatigue.

Here we aimed to test the effect of SJZD in a mouse model of depression with qi deficiency, and to investigate the underlying mechanisms, focusing on the inflammatory factors in both the central and

peripheral systems. By taking advantage of some shared genetic predisposition for depression and fatigue, we constructed a co-susceptibility model of depression with fatigue/qi-deficiency in a strain-dependent manner. Two strains of mice (Balb/c and 129S1/SvImJ (129/S1)) have been shown susceptibility to stress and vulnerable to depression (Rodriguez et al., 2020). Balb/c mice are frequently used for induction of qi-deficiency/fatigue phenotypes (Norden et al., 2015b), whereas we have previously observed that 129/S1 mice are resistant to qi-deficiency/fatigue. Therefore, these two strains of mice were compared to demonstrate the intrinsic difference in qi deficiency/fatigue behaviors, in spite of comparable susceptibility to depression, following the same procedure of chronic unpredictable mild stress (CUMS). The validity of the model was further verified by administration of SJZD to test its differential effects on the two strains of mice. FLX was used to be compared with SJZD behaviorally and molecularly, as previous studies suggest it has no effect on fatigue (Norden et al., 2015a). The inflammatory factors in both hippocampus and muscles were examined in depression with qi-deficiency mouse model and the following treatment with SJZD or FLX.

2. Materials and methods

2.1. Animals

Male Balb/c and 129S1/SvImJ (129/S1) mice, weight 20–22 g, were purchased from Shanghai Changzhou Cavince Laboratory Animal Co., Ltd (License number: SCXK (SU)-2016-0010). Mice were kept in a barrier room registered with the SPF and fed standard food and filtered water. The "Animal Experiment Protocol" complies with the "Guidelines for the Care and Use of Laboratory Animals" and was approved by the Animal Ethics and Use Committee of Jinan University (20230220-0031).

2.2. Drugs

Sijunzi decoction (SJZD) was consisted of *Radix Codonopsis* (200 g), *Atractylodis Macrocephalae Rhizoma* (200 g), *Poria* (200 g), *Glycyrrhizae Radix Et Rhizoma Praeparata Cum Melle* (200 g), which was purchased from Nanjing University of Chinese Medicine Chinese Pharmacopoeia, the outpatient department of the School of Medicine (License number: 20220301-01). After soaking in distilled water for 30 min, SJZD was boiled twice in distilled water, the first time at 1: 8 w/v for 30 min and the second time at 1:4 w/v for 30 min. The decoctions after the two boils were filtered, combined, and freeze-dried. The effective doses screening was shown in Supplementary Fig. 1, and we chose dosage of SJZD (5 g/kg/day) followed the 'Chinese Pharmacopoeia (2020)'. and has been used experimentally (Yu et al., 2016). Fluoxetine (Merck, CAS No.: 56296-78-7), a conventional SSRI antidepressant, was administrated with the dose 18 mg/kg/day, according to previous studies (Zou et al., 2017). Based on previous findings, fluoxetine, an SSRI antidepressant without influencing fatigue on the animal models (Carratalá-Ros et al., 2021; Norden et al., 2015a), was used as a negative control for anti-fatigue/qi deficiency.

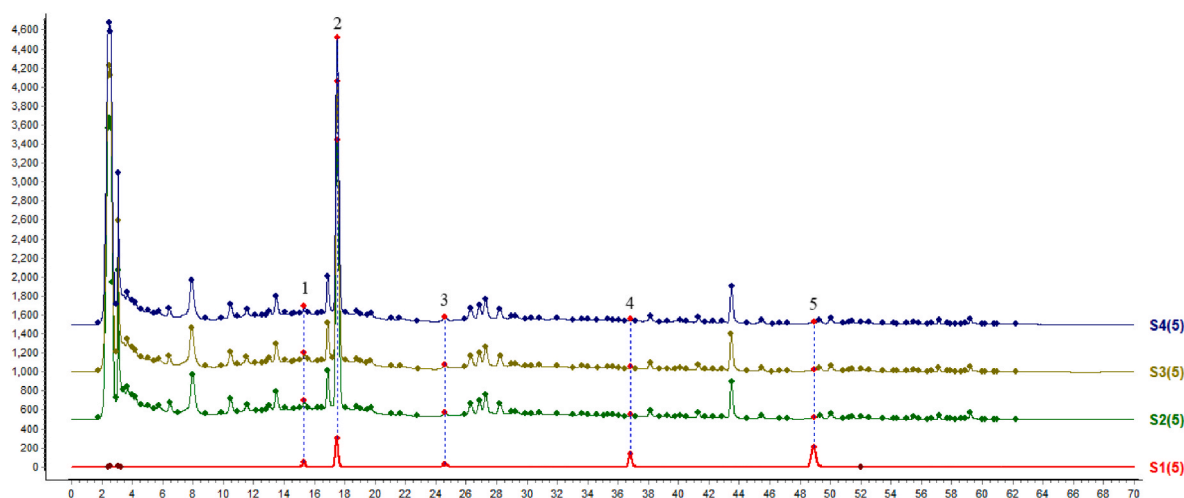


Fig. 1. The HPLC of sijunzi decoction. S1 is the reference solution, S2, S3, and S4 are three different batches of SJZD preparations. The peaks represent the following compounds: 1-Dangshen glycoside I, 2-liquiritigenin, 3-Dangshen glycoside, 4-Isorhamnetin, 5-Atractylodes lactone III.

Table 1
Contents of 5 compounds in SJZD.

Compounds	Repeat times (mg/g, n = 3)			Mean proportion (mg/ml)
	1	2	3	
Dangshen glycoside II	0.504	0.559	0.560	0.541
liquiritigenin	14.621	14.927	14.860	14.803
Dangshen glycoside	0.523	0.524	0.527	0.525
Isorhamnetin	0.022	0.021	0.021	0.021
Atractylodes lactone III	0.112	0.114	0.117	0.114

2.2.1. Quantitative analysis results of SJZD by HPLC

Preparation of reference substance solutions and sample solutions.

We accurately weighed 0.38 mg of Dangshen glycoside I, 1.06 mg of liquiritigenin, 1.33 mg of Dangshen glycoside, 1.33 mg of Isorhamnetin, and 4.76 mg of Atractylodes lactone III in separate 5 ml volumetric flasks. We dissolved each in methanol with ultrasound, allowed to cool to room temperature, diluted to volume with methanol, and filtered through a 0.22 μ m membrane to obtain the solutions. We then took 4 ml of the sample solution, added 4 ml of methanol, processed with ultrasound, centrifuged at 5000 r/min for 5 min, and collected the supernatant.

2.2.2. Chromatographic conditions and quantitative analysis

We used an Agilent Eclipse XDB-C18 column (4.6 mm $\times$ 250 mm, 5 μ m). The mobile phase consists of acetonitrile (A) and 0.05 % phosphoric acid water (B). The flow rate is 1.0 ml/min, detection wavelength is 220 nm, column temperature is 40 $^{\circ}$ C, and the injection volume is 10 μ L. We prepared sample solutions from 4 ml of the medicinal solution of three batches and calculated the content of Dangshen glycoside I, liquiritigenin, Dangshen glycoside, Isorhamnetin, Atractylodes lactone III in each sample. HPLC analysis was used for the quality control of the preparations of SJZD, shown in Fig. 1 and Table 1.

2.2.3. Experimental design

To establish the animal models, 129/S1 and Balb/c mice were exposed to Chronic unpredictable mild stress (CUMS) for 4 weeks, followed by behavioral tests for depression and qi deficiency. For the pharmacological studies, in each of the strain, animals are randomly assigned to four groups (non-stress Control, Vehicle, SJZD, FLX), the CUMS-exposed mice were administrated with drugs once a day for 1 week during the last week of CUMS exposure. Behavioral tests for depression and qi deficiency were carried out at 1 day after the last drug

treatment. The animals were then sacrificed for molecular study following the behavioral tests. The procedure for each experiment was shown in Figs. 3A, 4A and 5A and 6A.

2.3. Chronic unpredictable mild stress (CUMS) procedure

CUMS protocols are designed to maximize the unpredictability of stressors, which were used to induce both depression or fatigue previously (Dong et al., 2013). The procedure was followed with previous studies (Tang et al., 2015; Zhang et al., 2020). One of the following stressors was applied daily in an unpredictable random order for 4 weeks: food and water were deprived for 24 h, 45 $^{\circ}$ cage tilt for 24 h, cage shaking for 40 min (high-speed horizontal shaking, 200 transfers/min), 20 h cage dirty (200 ml water in sawdust pad), night illumination for 12 h, tube 50 ml for 6 h. We kept all CUMS mice individually in their cages until the end of the experiment. Non-stressed control mice were housed in the normal condition without stress administration.

2.4. Behavioral tests for depression

Sucrose preference test (SPT). SPT is used to assess the symptoms of anhedonia (Gross and Pinhasov, 2016). The procedure was according to our previous published articles (Tang et al., 2015; Zhang et al., 2020). Briefly, mice were conditioned for 3 days in double bottles of 2 % sucrose water, then deprived of water for 18 h, and tested in double bottles (one bottle each of water and sucrose water) for 6 h. We calculated the detection indicators of sucrose water consumption/sucrose water + water consumption. Weekly testing from the second week of molding was carried out.

Tail suspension test (TST). TST is a classic and rapid method for evaluating the efficacy of antidepressants, stimulants, and sedative drugs (Cryan et al., 2005). The protocol was according to our previous studies (Tang et al., 2015; Zhang et al., 2020). In brief, mouse tail is glued to the hook about 1 cm, hung upside down in the test box, tested for 6 min, and accumulated immobility time within the last 4 min after recording.

Forced swimming test (FST). FST is mainly used for the study of antidepressant, sedative, and analgesic drugs (Thierry et al., 1984). The protocol was according to our previous studies (Tang et al., 2015; Zhang et al., 2020). Mice were placed into a cylindrical glass jar 30 cm high and 18 cm in diameter, water depth in the tank was 10 cm, water temperature was 25 $\pm$ 1 $^{\circ}$ C. mice were forced to swim for 6 min, and the cumulative floating time of mice within the last 4 min after recording.

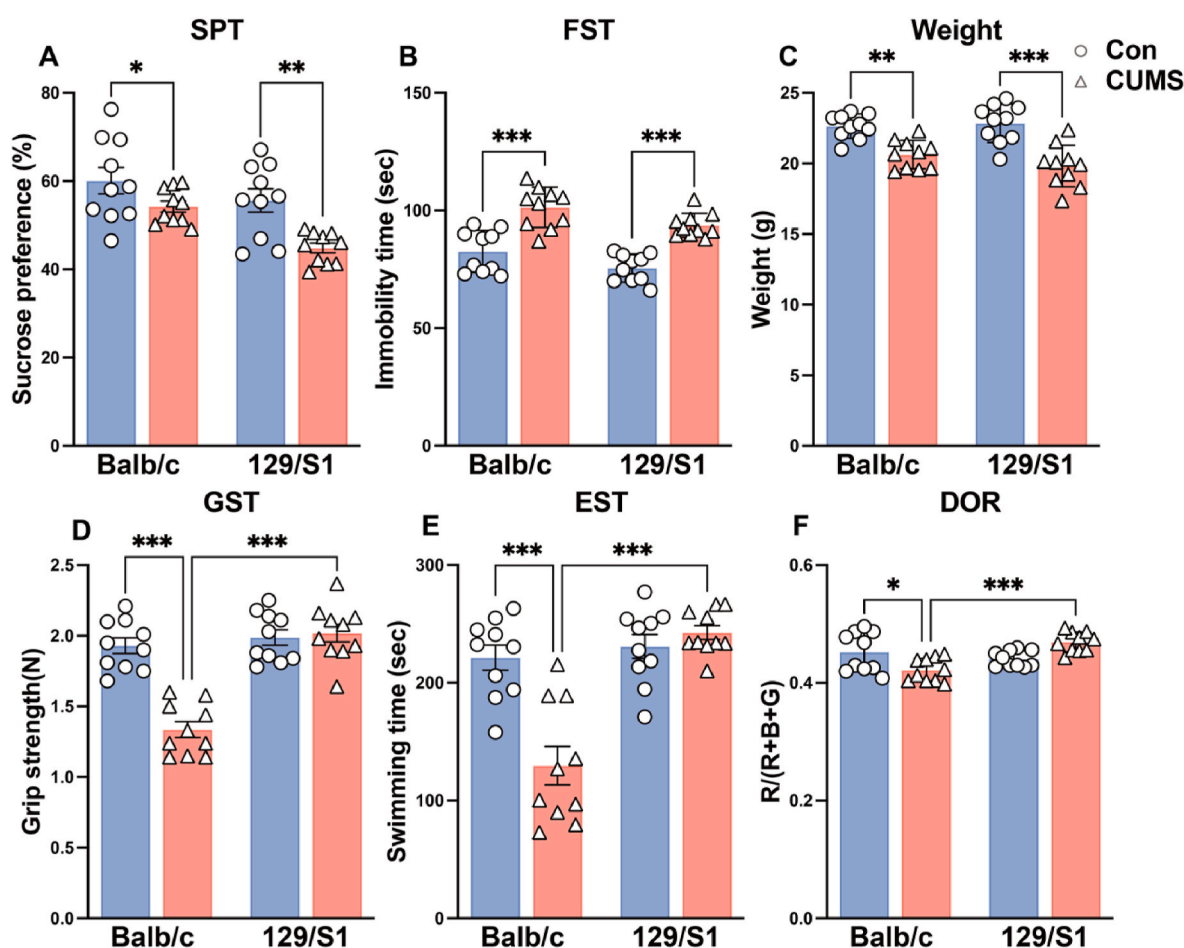


Fig. 2. Balb/c mice exhibited depression with qi deficiency phenotypes, but 129/S1 mice not. (A) Sucrose preference ratio in SPT. (B) Immobility time in FST. (C) Weight alteration after CUMS. (D) Grip strength in GST. (E) Exhaustive time in EST. (F) Degree of redness in DOR. Two-way NAOVA, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; $n = 10$ /group.

2.5. Behavioral tests for qi-deficiency

Grip strength test (GST). Qi deficiency was typically measured by grip strength in mice (Cho et al., 2018; Liu, L. et al., 2021). The grip of the mice was recorded with a handheld push/pull force meter (HP-100 digital). The procedure was repeated for 5 times, and the average score was used for each individual mouse. The score was normalized to the weight of the mouse. In this way, the grasp evaluation value of mice under natural response conditions was evaluated.

Exhaustive swimming test (EST). EST was used to evaluate the anti-fatigue effect of drugs (An et al., 2023; Liu, L. et al., 2021). Weight-bearing was applied at 8 % of the mouse's body weight and fixed at 1–3 cm of the mouse's tail. We placed mice in a clear glass beaker filled with about 3L of water at a depth of about 15 cm and kept the water temperature at 25 ± 1 °C. We observed and recorded the time from when the mouse was placed in the water to when the exhaustion sank to the bottom for more than 8 s.

Degree of redness (DOR). DOR is a measurement of blood deficiency, a common consequence of qi deficiency, and thus used as an indirect index for qi-deficiency (Liu, L. et al., 2021). At 25 °C room temperature 100lux illumination, mouse forepaw vein pictures were collected by stereo microscope magnification 20 times. We used Photoshop CS software to analyze the intermediate context of the second and third palms of the front paw. We evaluated the metric using the R-value divided by $R + G + B$.

Among these tests, GST is one of the mostly used index for evaluating qi deficiency. EST is also commonly used, but compounded with some

emotional components. DOR directly measured the blood deficiency, which is closely linked with qi deficiency, and thus used as an indirect and supplementary index for qi deficiency.

2.6. Quantitative real-time PCR analysis (qPCR)

RNA was isolated from the whole hippocampus (ventral and dorsal) and muscle (20 mg) using Trizol reagent (Invitrogen) and reverse-transcribed to cDNA using the SYBR PrimeScript RT-PCR kit (Takara). Quantitative RT-PCR of 1.5 μ l DNA was using SYBR green master mix reagent (Takara). The primer showed the following: IL-1 β forward: 5'-TGTGGCAGCTACCTATGTCT-3', reverse: 5'-GGGAACATCACACACTAGCA-3'; TNF- α forward: 5'-CATCTTCTCAAACTCGAGTGACAA-3', reverse: 5'-TGGGAGTAGATAAGGTACAGCCC-3'; NF-kB forward: 5'-CGGCGGAATTAACGTGGAC-3', reverse: 5'-CGAACCCGAGCTTGAGCTG-3'; CD8 forward: 5'-CGCAGCTCTCCTATGCAGTG-3', reverse: 5'-GTGTTTGGATCGTCAGGAAT-3'; GAPDH forward: 5'-CATGAGCAGTAACGCCATGAT-3', reverse: 5'-AGTACACCCATCGAATTCCAGT-3'. Fold changes in gene expression (encoding exons) were normalized to GAPDH. qPCR was performed according to the manufacturer's manual. The $\Delta\Delta$ CT method was used to obtain the relative expression values.

2.7. Data analysis

The treatment effect of the two samples was analyzed by using the student t-test, two-tailed. For three or more groups, one-way or two-way

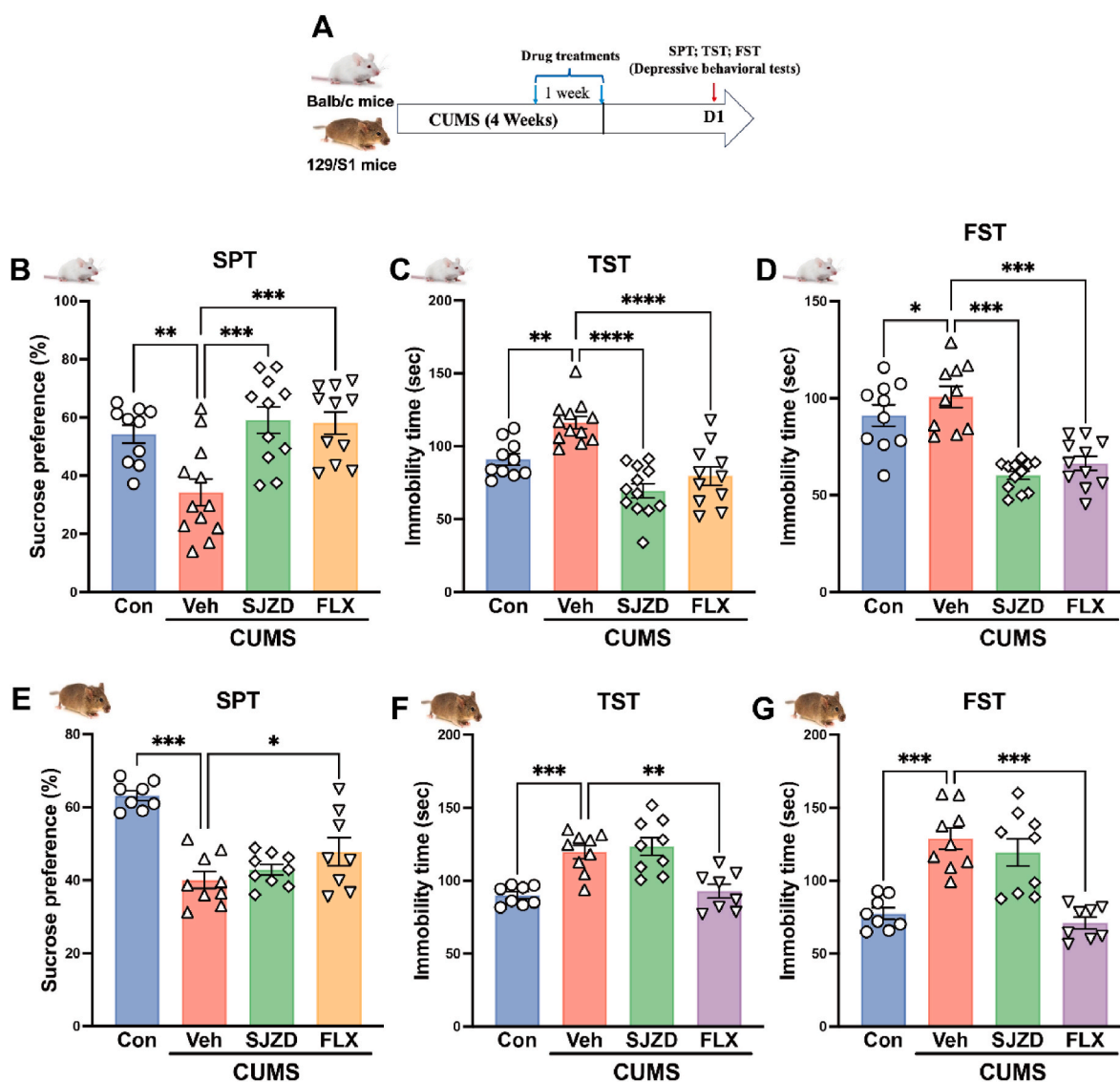


Fig. 3. SJZD improved depressive phenotypes in Balb/c mice or not 129/S1 mice. (A) The procedure of experimental design and drug treatments. (B) Sucrose preference ratio in SPT in Balb/c mice. (C) Immobility time in TST in Balb/c mice. (D) Immobility time in FST in Balb/c mice. (E) Sucrose preference ratio in SPT in 129/S1 mice. (F) Immobility time in TST in 129/S1 mice. (G) Immobility time in FST in 129/S1 mice. One-way ANOVA, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs Vehicle group; $n = 8$ –13/group.

analysis of variance is applied, followed by Tukey's post hoc test. All data were expressed as mean $\pm$ SEM, with statistical significance at the 5 % level acceptable.

3. Results

3.1. The main bioactive ingredients of SJZD

We analyzed the following 5 active ingredients to evaluate the stability of the SJZD: Dangshen glycoside I, liquiritigenin, Dangshen glycoside, Isorhamnetin, Atractylodes lactone III. The contents of these 5 components in the three batches of SJZD were very similar, indicating the stability and repeatability of the extraction process. The average concentrations of the 5 components in SJZD were as follows: 0.541 mg/ml, 14.803 mg/ml, 0.525 mg/ml, 0.021 mg/ml and 0.114 mg/ml, respectively.

3.2. Following CUMS, Balb/c and 129/S1 strain mice both exhibited depression with qi deficiency only shown in Balb/c mice

We found that SJZD had no antidepressant effect in the Balb/c mice without qi-deficiency under the normal condition (Supplementary Fig. 2). After exposure to the same procedure of chronic unpredictable mild stress on both Balb/c and 129/S1 strain mice, we performed behavioral tests for depression and qi deficiency. We found that both Balb/c and 129/S1 mice showed significant weight loss, reduced sucrose preference ratio in SPT, and increased immobility time in FST, indicating both Balb/c and 129/S1 mice exhibited significant depression-like phenotypes (Fig. 2A–C, Two-way ANOVA, SPT, $F(1, 36) = 1.263$, $P = 0.2685$; FST, $F(1, 36) = 0.01893$, $P = 0.8913$; weight, $F(1, 36) = 1.825$, $P = 0.1852$). However, qi deficiency phenotypes showed significant differences between Balb/c and 129/S1 mice. There was a significant reduction of grip in GST, swimming time in EST and DOR in the claws only in Balb/c mice, but not 129/S1 mice (Fig. 2D and E, Two-way ANOVA, GST, main effect interaction, $F(1, 36) = 29.36$, $P < 0.0001$, CUMS, $F(1, 36) = 41.27$, $P < 0.0001$, strain $\times$ CUMS, $F(1, 36) = 23.82$, $P < 0.0001$. EST, main effect interaction, $F(1, 36) = 20.80$, $P < 0.0001$,

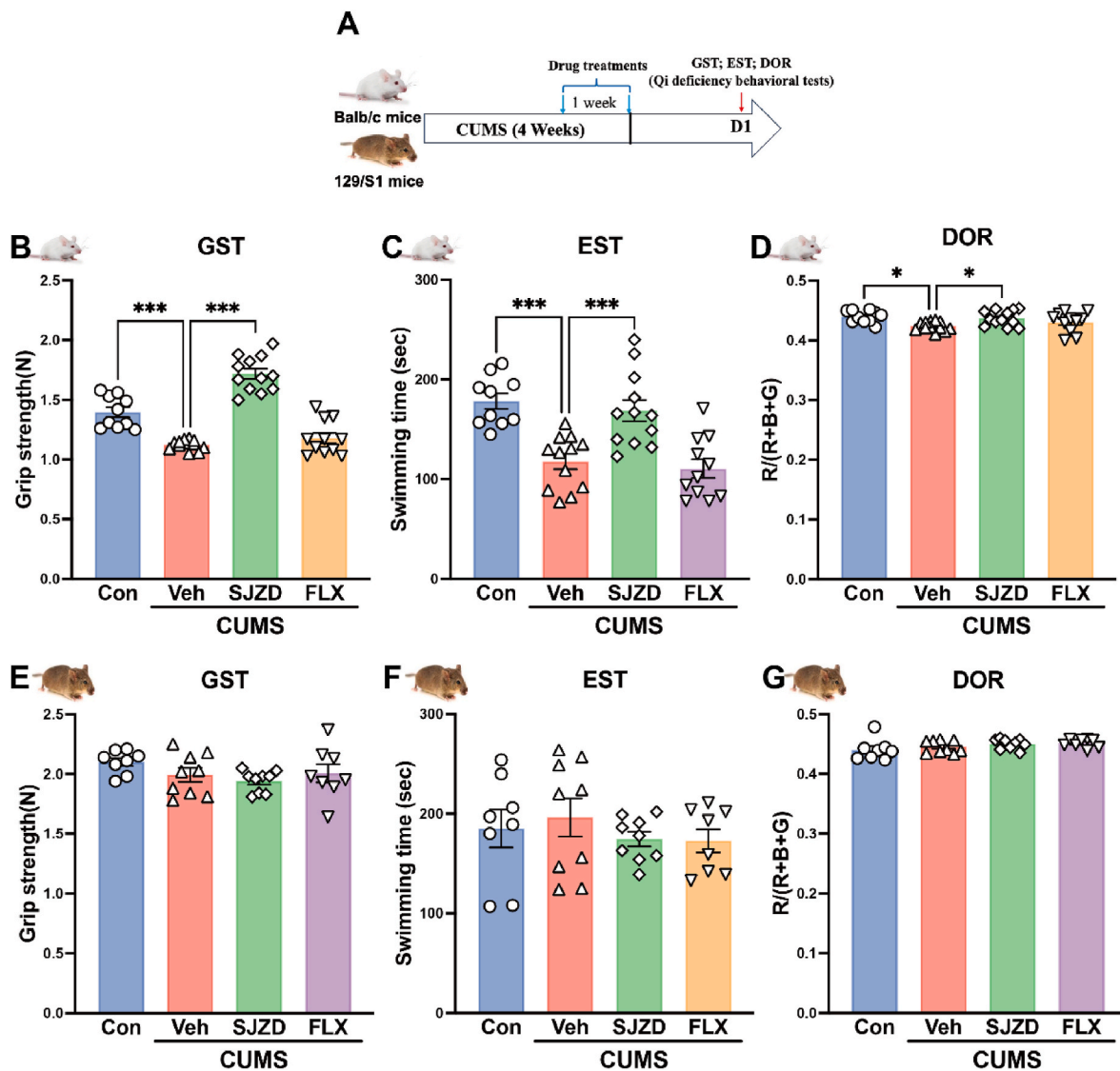


Fig. 4. SJZD improved Qi deficiency in Balb/c mice, but FLX not. (A) The procedure of experimental design and drug treatments. (B) Grip strength in GST in Balb/c mice. (C) Exhaustive time in EST in Balb/c mice. (D) Degree of redness in DOR in Balb/c mice. (E) Grip strength in GST in 129/S1 mice. (F) Exhaustive time in EST in 129/S1 mice. (G) Degree of redness in DOR in 129/S1 mice. One-way ANOVA, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs vehicle group; $n = 8-13$ /group.

CUMS, $F(1, 36) = 29.29$, $P < 0.0001$, strain $\times$ CUMS, $F(1, 36) = 12.45$, $P = 0.0012$. DOR, main effect interaction, $F(1, 36) = 17.60$, $P = 0.0002$, CUMS, $F(1, 36) = 6.609$, $P = 0.0144$, strain $\times$ CUMS, $F(1, 36) = 0.01949$, $P = 0.018$. These data demonstrated that depression with qi deficiency was only induced in Balb/c mice, but not 129/S1 mice, following the CUMS procedure.

3.3. SJZD improved CUMS-induced depression only in the Balb/c mice, but not 129/S1 mice

We then evaluated the validity of SJZD on depression with qi-deficiency model in Balb/c strain mice. The procedure of experimental design and drug treatments was shown in Figs. 3A and 4A. We first carried out the effective dose screening experiment to verify the effectiveness of three doses (2.5, 5, 7.5 g/kg), and the results showed that all three doses improved the immobility time of TST in mice, but only the doses of 5 and 7.5 g/kg improved the grip value of GST in mice, as shown in Supplementary Fig. 1. Therefore, we chose 5 g/kg for experimental verification. In the depressive behavioral tests, we found that CUMS significantly reduced the sucrose preference in SPT and increased immobility time both in TST and FST, which were all reversed by SJZD

after 7 days treatments in Balb/c strain mice, but not 129/S1 strain mice, whereas FLX reversed them in both strains (Fig. 3B–G, One-way ANOVA, Balb/c mice, SPT, $F(3, 40) = 8.374$, $P = 0.0002$; TST, $F(3, 41) = 17.50$, $P < 0.0001$; FST, $F(3, 41) = 18.06$, $P < 0.0001$. 129/S1 mice, SPT, $F(3, 30) = 18.10$, $P < 0.0001$; TST, $F(3, 30) = 13.65$, $P < 0.0001$; FST, $F(3, 30) = 18.47$, $P < 0.0001$).

3.4. SJZD, but FLX, improved CUMS-induced qi deficiency in the Balb/c mice

The effects of SJZD on qi deficiency behavioral tests including GST, EST and DOR were also compared. The grip strengthening in GST, swimming time in EST and claw color in DOR were all obviously decreased by CUMS only in Balb/c strain mice, but not 129/S1 strain mice. SJZD relieved all behavioral phenotypes of qi deficiency, which were not affected by FLX (Fig. 4B–G, One-way ANOVA, Balb/c mice, GST, $F(3, 41) = 39.19$, $P < 0.0001$; EST, $F(3, 41) = 14.08$, $P < 0.0001$; DOR, $F(3, 41) = 8.117$, $P = 0.0002$. 129/S1 mice, GST, $F(3, 30) = 1.622$, $P = 0.2051$; EST, $F(3, 30) = 0.5248$, $P = 0.6686$; DOR, $F(3, 30) = 1.583$, $P = 0.2140$). These data indicated that SJZD was capable of improving the symptoms of depression and qi deficiency only in Balb/c

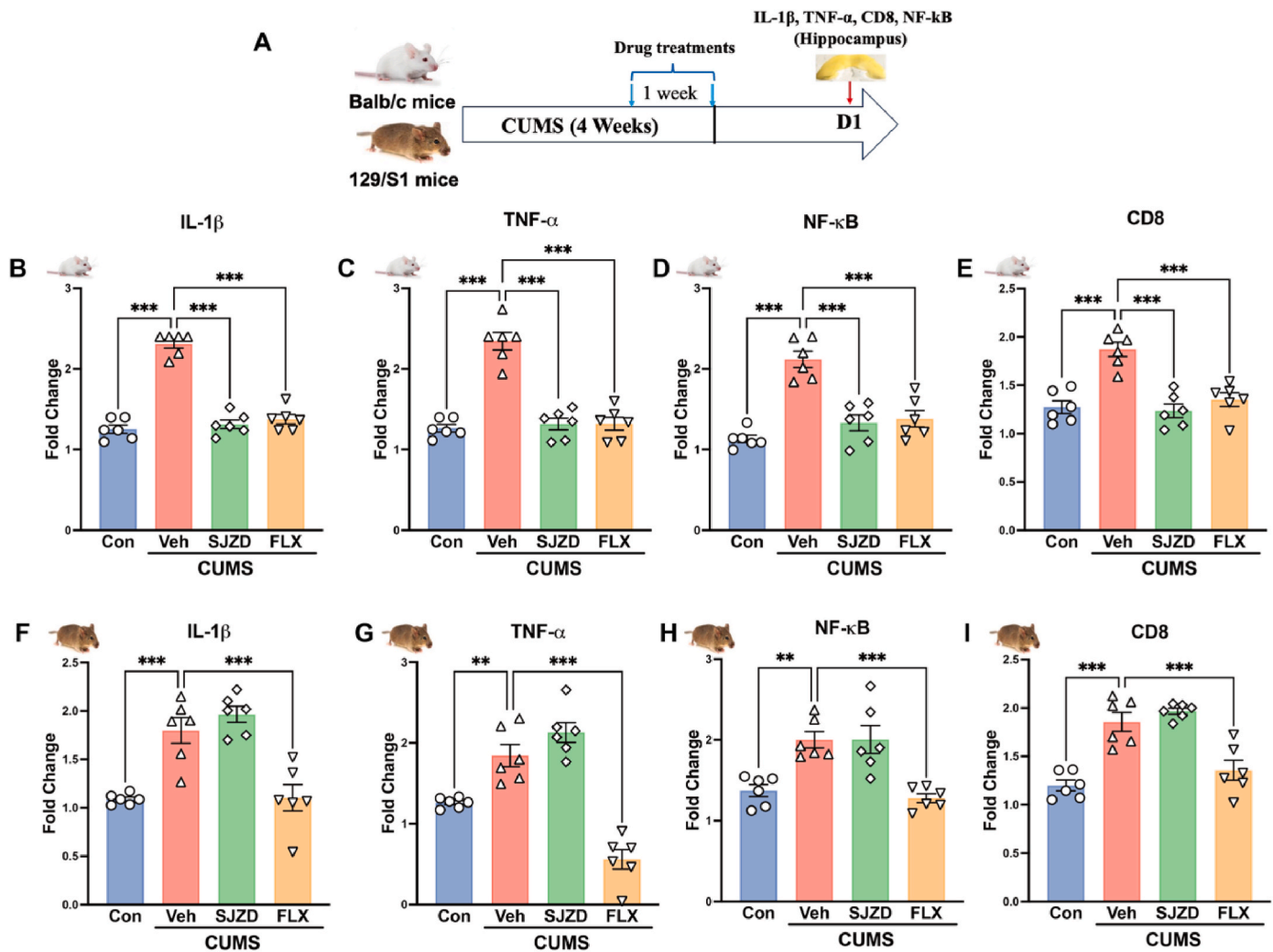


Fig. 5. SJZD improved inflammatory factors' expressions in the hippocampus in Balb/c mice but not 129/S1 mice. (A) The procedure of experimental design and drug treatments. (B) The expression of IL-1 β in Balb/c mice. (C) The expression of NF- κ B in Balb/c mice. (D) The expression of TNF- α in Balb/c mice. (E) The expression of CD8 in Balb/c mice. (F) The expression of IL-1 β in 129/S1 mice. (G) The expression of NF- κ B in 129/S1 mice. (H) The expression of TNF- α in 129/S1 mice. (I) The expression of CD8 in 129/S1 mice. One-way ANOVA, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vehicle group; $n = 6$ /group.

mice, whereas FLX could improve depressive symptoms in both strains, without influencing qi deficiency in Balb/c mice.

3.5. SJZD suppressed expressions of inflammatory factors in the hippocampus of Balb/c mice, but not 129/S1 mice

In the hippocampus, the gene expressions of IL-1 β , TNF- α , CD8 and NF- κ B in both Balb/c and 129/S1 mice were significantly increased by CUMS. SJZD reversed them in Balb/c mice only, without affecting them in 129/S1 mice. In contrast, FLX reversed them in both mouse strains (Fig. 5B–I, One-way ANOVA, Balb/c mice, IL-1 β , $F(3, 20) = 83.38$, $P < 0.0001$; TNF- α , $F(3, 20) = 42.19$, $P < 0.0001$; NF- κ B, $F(3, 20) = 23.21$, $P < 0.0001$; CD8, $F(3, 20) = 17.73$, $P < 0.0001$. 129/S1 mice, IL-1 β , $F(3, 20) = 19.27$, $P < 0.0001$; TNF- α , $F(3, 20) = 38.89$, $P < 0.0001$; NF- κ B, $F(3, 20) = 12.67$, $P < 0.0001$; CD8, $F(3, 20) = 23.01$, $P < 0.0001$).

3.6. SJZD, but not FLX, suppressed expressions of inflammatory factors in the muscles of Balb/c mice

In the muscles, the gene expressions of IL-1 β , TNF- α , CD8 and protein expression of NF- κ B both in Balb/c and 129/S1 mice after SJZD treatments for 7 days were examined. The procedure was shown in Fig. 6A. Gene expressions of IL-1 β , TNF- α , CD8 and NF- κ B were all significantly

increased in CUMS-exposed Balb/c mice, which were all reversed by SJZD, but not by FLX. In contrast, they were not changed either by CUMS or by treatment with SJZD or FLX in 129/S1 mice (Fig. 6B–I, One-way ANOVA, Balb/c mice, IL-1 β , $F(3, 12) = 15.79$, $P = 0.0002$; TNF- α , $F(3, 12) = 3.210$, $P = 0.018$; NF- κ B, $F(3, 12) = 17.23$, $P = 0.0001$; CD8, $F(3, 12) = 5.719$, $P = 0.0115$. 129/S1 mice, IL-1 β , $F(3, 12) = 2.353$, $P = 0.1235$; TNF- α , $F(3, 12) = 2.179$, $P = 0.1434$; NF- κ B, $F(3, 12) = 0.7036$, $P = 0.5679$; CD8, $F(3, 12) = 0.5225$, $P = 0.6749$).

4. Discussion

The present study aimed to study the effects and mechanism of SJZD in the treatment of depression with qi-deficiency by using a valid animal model. We showed that Balb/c strain mice conferred co-susceptibility to qi deficiency and depression after exposed to CUMS, in contrast to the single susceptibility to depression for 129/S1 strain mice. SJZD was able to reversed depressive behaviors and qi deficiency selectively in CUMS-exposed Balb/c mice, but not 129/S1 mice. SJZD down-regulated expressions of inflammatory factors in the hippocampus as well as muscles in CUMS-exposed Balb/c mice. These findings suggested that SJZD stably improved both depression and qi deficiency in CUMS-exposed Balb/c mice, which was involved in its anti-inflammation effects in the muscles and hippocampus.

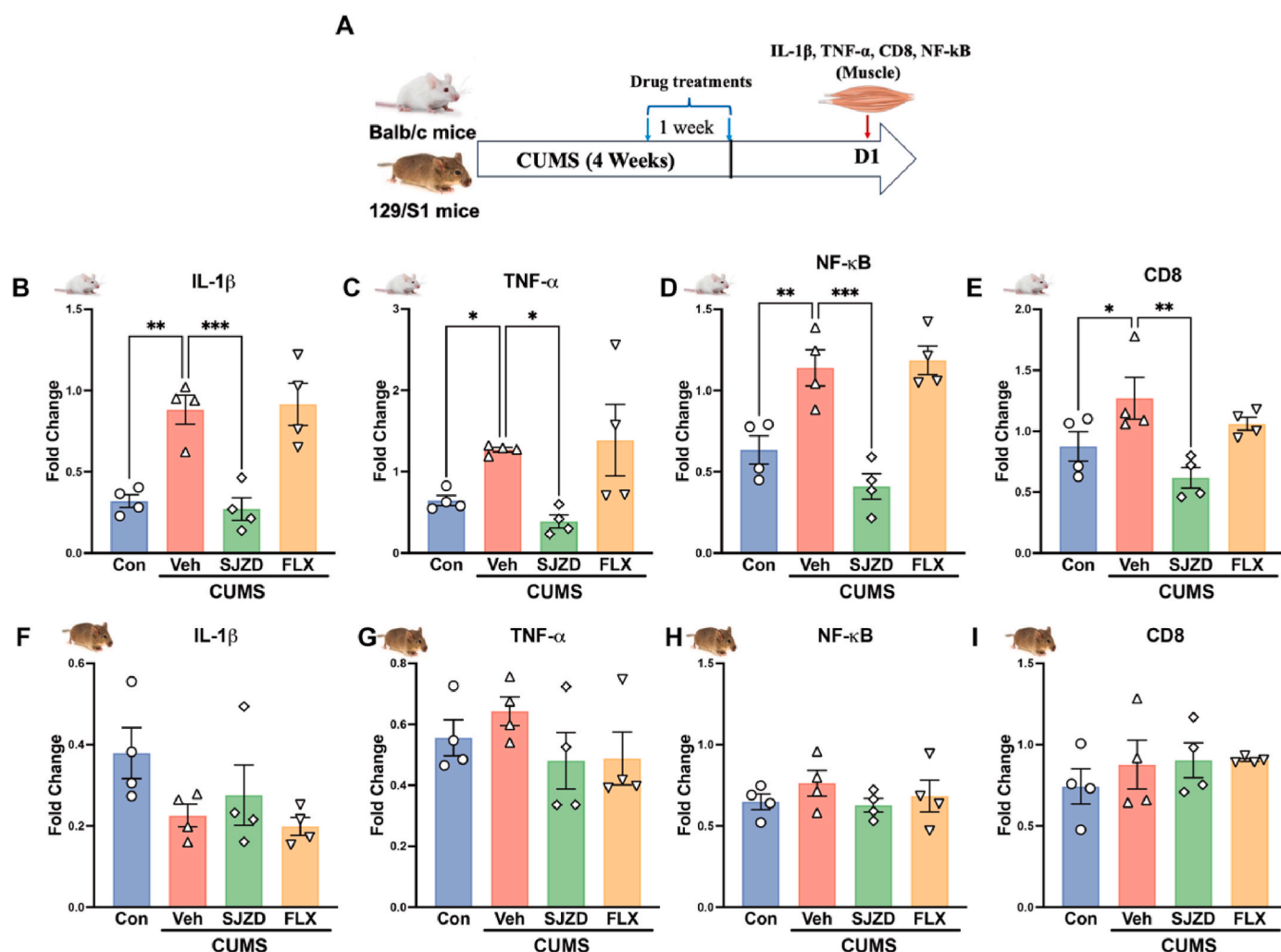


Fig. 6. SJZD improved inflammatory factors' expressions in the muscle only in Balb/c mice. (A) The procedure of experimental design and drug treatments. (B) The expression of IL-1 β in Balb/c mice. (C) The expression of NF- κ B in Balb/c mice. (D) The expression of TNF- α in Balb/c mice. (E) The expression of CD8 in Balb/c mice. (F) The expression of IL-1 β in 129/S1 mice. (G) The expression of NF- κ B in 129/S1 mice. (H) The expression of TNF- α in 129/S1 mice. (I) The expression of CD8 in 129/S1 mice. One-way ANOVA, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vehicle group; $n = 4$ /group.

The present study, based on the differences in susceptibility to both qi deficiency syndrome and depression, constructed a stable model of depression accompanied by qi deficiency following CUMS exposure. Depression is a hetero-generous disease, caused by different genetic and environmental factors, which may explain why there were the different effects by using the same type of drug treatment. Treating depression accordingly in terms of the different subtypes, or stratified treatment, represented an important direction to improve the efficacy (Lan et al., 2024; Sun et al., 2024). For the same individual disease including depression, there are different TCM syndrome (Li et al., 2020), which is also at least partially determined by genetic factors. This made it feasible to identify the genetic-based model for co-susceptibility to depression and qi deficiency, as demonstrated here by the Balb/c strain mice. In contrast, 129S1 mice was comparably susceptible to depression without showing qi deficiency. The following evidence support to use of chronically stressed Balb/c mice as a model of depression with qi deficiency: First, only Balb/c mice showed qi deficiency phenotypes following CUMS, and only SJZD, a qi-tonifying formula, but not the conventional antidepressant FLX, improved the qi deficiency on Balb/c mice. Secondly, SJZD improved depression only in Balb/c mice showing qi-deficiency following CUMS. It had no antidepressant effect in the Balb/c mice without qi-deficiency under the normal condition (Supplementary Fig. 2), and it had no antidepressant effects on 129S1

mice, which had no qi deficiency. In contrast, FLX had antidepressant effects on both strains, without requirements of qi-deficiency. Finally, qi-deficiency associated peripheral muscle inflammation was only found in CUMS-exposed Balb/c mice, and could only be attenuated by SJZD. Balb/c mice as a co-susceptible strain was of great significance for the research and development of treatment of depression with qi-deficiency or fatigue. Studies have shown that both fatigue and depression are genetically influenced, and that certain common genetic factors may contribute to their co-occurrence (Keller, 2018; Slyepchenko et al., 2017). Therefore, our findings support the validity of Balb/c as a stable genetic model of depression with qi deficiency/fatigue.

The current study demonstrated that the qi tonifying formula SJZD alone was capable to elicit antidepressant effects, with the presence of qi deficiency. SJZD is widely used in clinical practice to treat qi deficiency syndrome, including fatigue, lazy speech and other symptoms (Ma et al., 2023). Long-term treatment of SJZD could improve limb weakness in rats (Liu, W.J. et al., 2021). SJZD also improved chronic atrophic gastritis accompanied with chronic fatigue (Dong et al., 2024; Huang and Shao, 2024; Li et al., 2024). On the basis of SJZD, there are several derivative formulae that deal with syndrome in addition to qi deficiency. For example, liu-jun-zi-decortion, or rikkunshito in kempo medicine, was able to reverse atrophy of myocardial and muscle fiber by potentiating ghrelin signaling (Fujitsuka et al., 2016). These findings support

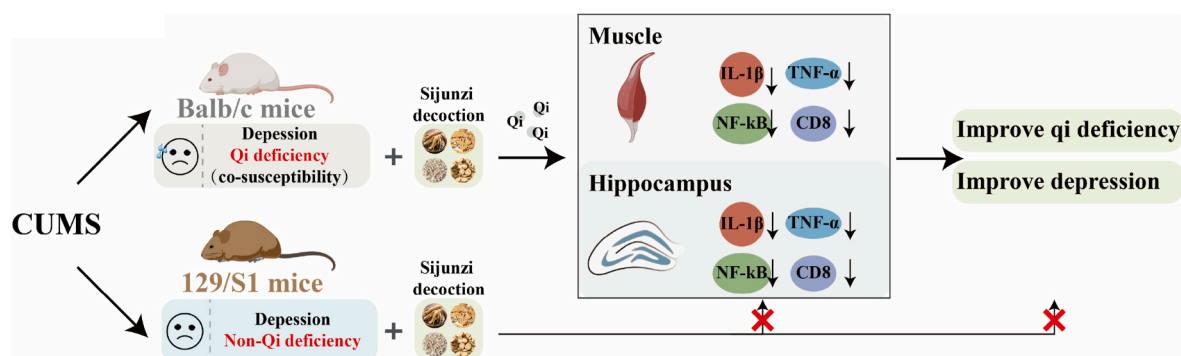


Fig. 7. Balb/c mice had predisposition to depression with qi deficiency, and Sijunzi decoction alleviated depression with qi deficiency/fatigue phenotypes, which required suppressing the activation of inflammation in the muscles and hippocampus. In contrast, 129/S1 strain mice showed depression without qi deficiency/fatigue, and were non-responsive to Sijunzi treatment behaviorally and molecularly.

that SJZD has broad beneficial effects on different systems in the body, in degenerating and inflammation-affected organs. In contrast, FLX was capable of relieving depressive symptoms both in Balb/c and 129/S1 mice exposed to CUMS, but showing little improvement in qi deficiency in both strains of mice. These findings suggest that qi deficiency may be one contributing factor for some subtype of depression, and SJZD improvement on qi deficiency may subsequently lead to therapeutic effects on depression, whereas FLX deals more directly with the depression symptoms. Importantly, our findings provide the first scientific evidence in terms of animal genetics to support the necessity and efficacy of stratified treatment of diseases like depression, using TCM treatment based on syndrome differentiation.

Up to now, the mechanism underlying co-occurrence of depression and fatigue or qi deficiency remains largely unknown. The present study for the first time demonstrated that it is closely associated with inflammation in both brain and peripheral systems. Consistent with previous studies, we found the activation of inflammation in the hippocampus in CUMS-exposed Balb/c and 129/S1 strain, regardless of the co-occurrence of qi deficiency (Nikolopoulos et al., 2023). However, the un-regulated expressions of IL-1 β , TNF- α , CD8 and NF- κ B inflammatory factors/signaling in the muscles were only found in Balb/c mice with qi deficiency. SJZD, widely used to treat muscle atrophy (Ohbuchi et al., 2015; Terawaki et al., 2017), has been shown to prevent the occurrence and development of abnormal muscle function (Wang et al., 2023). In agreement with it, here we found SJZD effectively reduced the inflammation in the muscles only in the Balb/c mice with qi-deficiency. Our results demonstrated that, unlike the strain-dependence in the change of inflammation in the muscles, the inflammatory factors were impaired in both strains following CUMS in the hippocampus, suggesting that this signaling in the hippocampus was more closely associated with depression, but not qi deficiency phenotype. Consistently, FLX could improve the signaling in both strains, and SJZD improved it only in Balb/c mice, in agreement with their improvement effects on depression. It has been well documented that the enhancement of inflammatory factors in hippocampus contribute to depressive symptoms (Yuan et al., 2023). Muscles release a variety of hormones or other molecules, including adiponectin, which involve in regulation of depression (Liu et al., 2024). Inflammation in the muscles may not only result in qi-deficiency/fatigue, but also impair the regulatory signaling for producing the molecules to combat depression. This possibility warrants further investigation. Nonetheless, our current results suggested that the antidepressant effects of SJZD were, at least partially, due to its attenuation of qi deficiency by downregulating inflammation in the peripheral organ of muscle. Our findings of the potential pathophysiological role of the immune systems to fatigue or qi deficiency to depression shed light on improving the treatment of such subtype of depression.

We assayed the content of five active components in SJZD, namely Dangshen glycoside I, liquiritigenin, Dangshen glycoside, Isorhamnetin,

and Atractylodes lactone III. Their consistent levels across the three batches of extract indicate the extraction process was stable and repeatable. For instance, the content of Dangshen glycoside I and Dangshen glycoside showed little variation among different batches, confirming the reliability of the extraction process. These compounds may, at least partly, contribute to the foundation of SJZD's efficacy. For example, Isorhamnetin is known for its antioxidant properties (Güven et al., 2023), and Atractylodes lactone III has anti-inflammatory effects (Zhou et al., 2021). Future research should delve deeper into the specific contributions of each compound and the potential synergism of them, as well as broaden the analysis to include other potential active components like polysaccharides.

This study systematically, quantitatively and objectively revealed the depression and qi deficiency co-susceptibility following chronic stress in the Balb/c strain mice, in comparison to the 129/S1 strain mice, which are susceptible to depression, but resistant to qi deficiency. The depression with qi deficiency model was further validated using intervention of SJZD and further revealed the relationship between the TCM syndrome and the disease. Importantly, via the multisystem investigation, the study disclosed the characteristics and association of the disease and the syndrome at the molecular level. This research is innovative in the following three aspects: Firstly, we established a reliable model for depression with qi deficiency, using mouse strain comparison approach on the basis of the genetic predisposition. Secondly, we demonstrated the qi-tonifying formula SJZD selectively improved depression and qi deficiency selectively in Balb/c strain mice, which supports the validity of the model. Finally, we provided evidence on the inflammation in the brain and periphery to support, at the molecular levels, the plausible substrate underlying the effects of SJZD on both depression and qi deficiency. However, there are still shortcomings in this study, such as not strengthening the specific indicators of "qi deficiency", like "low energy metabolism, mitochondrial dysfunction", which will be further addressed in future studies. Our molecular findings for the current study are basically associative, but lay the foundation for further disclosing in-depth causal relationship by the future studies.

5. Conclusions

Balb/c strain mice showed predisposition to develop depression with qi deficiency following chronic stress, and Sijunzi decoction alleviated both phenotypes, which was associated with suppression of inflammation in the hippocampus and muscles, respectively (Fig. 7). This is the first time to demonstrate an animal model of depression with qi deficiency on the basis of strain, and the advantages of Sijunzi decoction in treatment of depression with qi deficiency. Our study offers an insight for developing a new therapeutic strategy for treatment of depression with qi deficiency or fatigue, and also elucidates the scientific connotation of stratified treatment of depression based on syndrome

differentiation of TCM.

CRediT authorship contribution statement

Jingwen Tan: Writing – original draft, Methodology, Investigation, Conceptualization. **Zhenzhen Fu:** Writing – original draft, Methodology, Investigation, Conceptualization. **Mingzhi Han:** Writing – original draft, Methodology, Investigation. **Nga-Lee Wong:** Methodology, Investigation. **Chenghao Song:** Methodology, Investigation. **Yin Chen:** Methodology, Investigation. **Wenjia Li:** Methodology, Investigation. **Maria A. Deli:** Writing – review & editing. **Hailou Zhang:** Writing – review & editing, Supervision, Investigation, Formal analysis, Conceptualization. **Gang Chen:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare there is no commercial or financial relationships that could be construed as a potential conflict of interest regarding this paper.

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Abbreviations

SJZD	Sijunzi decoction
MDD	major depressive disorder
TCM	traditional Chinese medicine
FLX	fluoxetine
CUMS	chronic unpredictable mild stress
TST	tail suspension test
FST	forced swimming test
SPT	sucrose preference test
NSF	novelty suppressed feeding test
GST	grip strength test
EST	exhaustive swimming test
DOR	degree of redness

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2025.120380>.

Data availability

The authors do not have permission to share data.

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